

Homework 1

Name:

Class:

Student ID:

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In linear algebra, the *concepts* are as important as the *computations*. The simple numerical exercises only help you check your understanding of basic procedures. Later in your career, **computers will do the calculations**, but you will have to **choose** the calculations, know how to **interpret** the results, and then **explain** the results to other people.

You must avoid the temptation to look at answers before you have tried to write out the solution yourself. **Otherwise, you are likely to think you understand something when in fact you do not.**

—*Linear Algebra and Its Applications*, by David C. Lay et al.

术语表 (Glossary)

英文术语	中文翻译	英文术语	中文翻译
Vector	向量	Zero vector	零向量
Vector space	向量空间	Subspace	子空间
Empty set	空集	Nonempty set	非空集
Plane	平面	Scalar	标量
Closed under addition	对加法封闭	Closed under scalar multiplication	对数乘（标量乘法）封闭
Union	并集	Intersection	交集
Linearly independent	线性无关	Linearly dependent	线性相关
Span	张成	Spanning (Generating) set	张成/生成集
Basis	基	Linear combination	线性组合
Dimension	维数	Finite-dimensional vector space	有限维向量空间
Sum of subspaces	子空间的和	Direct sum ($\oplus$)	直和
System of vectors	向量组	Polynomial	多项式
Quadrant	象限	Polynomial space ($\mathcal{P}_n(\mathbb{F})$)	($\mathbb{F}$ 上的 n 次) 多项式空间

常见证明术语

英文术语	中文解释
Since A , (we have) B	因为 A , 所以 B (注意没有 so)
Therefore, A	因此, A (常用于证明末尾)
There exists A such that B	存在 A 满足 B
which implies ...	这意味着……
which gives / yields	上式给出 (由上式可得) ……
Thus ...	则…… (表示因果关系, 一般放在段中)

还有很多讲述证明的方式，可以学习讲义。尽量让语言生动而清晰，不要用 $\therefore$ 和 $\therefore$ ，也不要从头 because/so 到尾，并且留心每个因果关系是否正确。善用行间公式，不要把所有的式子挤在一行。

1. Justify each answer. Draw $\checkmark$ if True and $\times$ if False.

- (a) The empty set is not a vector space. ()
- (b) A nonempty set that is closed under addition and scalar multiplication is a vector space. ()
- (c) A plane in $\mathbb{R}^3$ is a two-dimensional subspace. ()
- (d) Suppose U_1 and U_2 are subspaces of V . The intersection of U_1 and U_2 is a subspace of V . ()
- (e) $\{(x, y, z) \in \mathbb{C}^3 : x^3 = y^3\}$ is a subspace of $\mathbb{C}^3$. ()
- (f) A single vector by itself is linearly dependent. ()
- (g) If $H = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$, then $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is a basis for H . ()
- (h) A basis is a linearly independent set that is as large as possible. ()
- (i) If $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ and $\{\mathbf{w}_1, \dots, \mathbf{w}_m\}$ are linearly independent lists of vectors in V , then $\{\mathbf{v}_1 + \mathbf{w}_1, \dots, \mathbf{v}_m + \mathbf{w}_m\}$ is linearly independent. ()
- (j) V is a nonzero finite-dimensional vector space, and the vectors listed belong to V .
 - i. If there exists a set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ that spans V , then $\dim V \leq p$. ()
 - ii. If there exists a linearly independent set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ in V , then $\dim V \geq p$. ()
 - iii. If $\dim V = p$, then there exists a spanning set of $p + 1$ vectors in V . ()
 - iv. If there exists a linearly dependent set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ in V , then $\dim V \leq p$. ()
 - v. If every set of p elements in V fails to span V , then $\dim V > p$. ()
 - vi. If $p \geq 2$ and $\dim V = p$, then every set of $p - 1$ nonzero vectors is linearly independent. ()

2. Let W be the union of the first and third quadrants in the xy -plane. That is, let $W = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : xy \geq 0, x, y \in \mathbb{R} \right\}$.

- (a) If $\mathbf{u}$ is in W and c is any scalar, is $c\mathbf{u}$ in W ? Why?
- (b) If $\mathbf{u}, \mathbf{v} \in W$, is $\mathbf{u} + \mathbf{v}$ in W ? Why?
- (c) Is W a vector space?

3. Let X and Y be subspaces of a vector space V .

(a) Let $\mathbf{v} \in V$, $\mathbf{v} \notin X$. Prove that if $\mathbf{x} \in X$ then $\mathbf{x} + \mathbf{v} \notin X$.

(b) Use (a) to show that $X \cup Y$ is a subspace if and only if $X \subset Y$ or $Y \subset X$.

(c) Give an example that X and Y are both subspaces of a vector space V , but $X \cup Y$ is not.

Note: *The union of subspaces is usually not a subspace, which is the reason for us to consider the sum of subspaces.*

4. Prove or give a counterexample: If U_1 , U_2 , W are subspaces of V such that $U_1 \oplus W = U_2 \oplus W$, then $U_1 = U_2$.

5. Suppose that V is a finite dimensional vector space. Let a system of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r \in V$ be linearly independent but not generating. Show that it is possible to find a vector $\mathbf{v}_{r+1} \in V$ such that the system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r, \mathbf{v}_{r+1}$ is linearly independent. (Hint: Take for $\mathbf{v}_{r+1}$ any vector that cannot be represented as a linear combination $\sum_{k=1}^r \alpha_k \mathbf{v}_k$ and show that the system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r, \mathbf{v}_{r+1}$ is linearly independent.)

6. Let $U = \{p \in \mathcal{P}_4(\mathbb{F}) : p(2) = p(5)\}$, $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$.

- (a) Give the standard basis of $\mathcal{P}_4(\mathbb{F})$ and verify that it is indeed a basis. What is the dimension of $\mathcal{P}_4(\mathbb{F})$? What is the dimension of $\mathbb{F}^5$? (Hint: Can you see the relationship between them?)
- (b) Find a basis of U . (Hint: Try to answer: Is $1 \in U$? Is $(x - 2)(x - 5) \in U$? Are there any other “independent” vectors? What is the possible size of the linearly independent set?)
- (c) Find a subspace W of $\mathcal{P}_4(\mathbb{F})$ such that $\mathcal{P}_4(\mathbb{F}) = U \oplus W$. (Hint: The proof of Theorem 2.6.3 might help.)